Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-4 | Winter-2024

Course Code	:	2105CS412	**Date**	:	19-11-2024
Course Name	:	Computer Networks	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define computer network with applications.	**4**
	(B)	Explain types of computer network. (LAN, MAN and WAN)	**3**
		OR	
		Define topology. Explain any one in details.	
	(C)	Draw the OSI Reference Model.	**7**
		OR	
		Draw the TCP/IP Model.	
Q.2	**(A)**	Define Transmission Media. Classify Transmission Media.	**4**
	(B)	Explain the Twisted-pair cable in detail.	**3**
		OR	
		Explain the fiber optic cable in detail.	
	(C)	Discuss the unguided media with its types.	**7**
		OR	
		Compare guided media and unguided media.	
Q.3	**(A)**	Describe parity check technique with example.	**4**
	(B)	Explain Go-back N protocol with example.	**3**
		OR	
		Explain selective repeat protocol with example.	
	(C)	Describe checksum technique with example.	**7**
		OR	
		Describe CRC technique with example.	

Q.4 **(A)** Define routing. Explain types of routing. **4**

(B) Given IP address 169.25.45.40/20. Find below details: **3**
a) Class
b) Subnet mask
c) What are the total subnets and hosts?

OR

Given IP address 192.168.10.10/25. Find below details:
a) Class
b) Subnet mask
c) What are the total subnets and hosts?

(C) Describe IPv4 datagram with diagram. **7**

OR

Describe link state routing protocol.

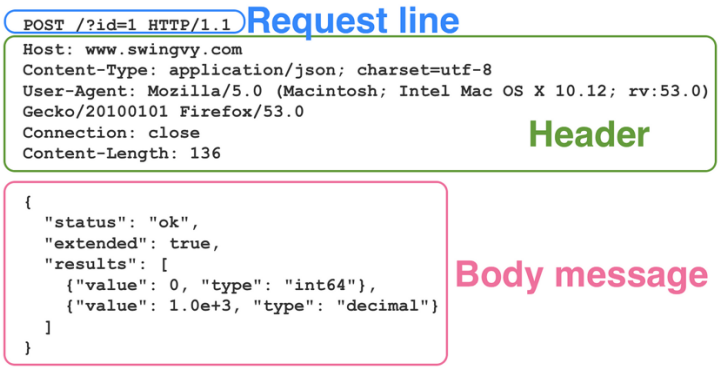
Q.5 **(A)** Explain DNS with example. **4**

(B) Draw TCP segment structure. **3**

OR

Draw UDP structure.

(C) Identify status code, URL, Browser name, Operating System used in HTTP message. **7**



OR

Draw rdt 2.0 with and without error in single FSM.

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